SECTION 3 · SUBSECTION 3.5
CODE RETRIEVAL

3.5.1 INTRODUCTION.

The retrieval stage builds a semantic index of the repository. It does not split text every N tokens; it emits one chunk for the module header, one for each function and one for each class, preserving code boundaries. Chunks are embedded by a hosted NVIDIA model and written to a ChromaDB collection scoped to the job. The one idea the reader must leave with is that stage 6 never reads whole files from GitHub; it reads these chunks (figure 3.5-1).

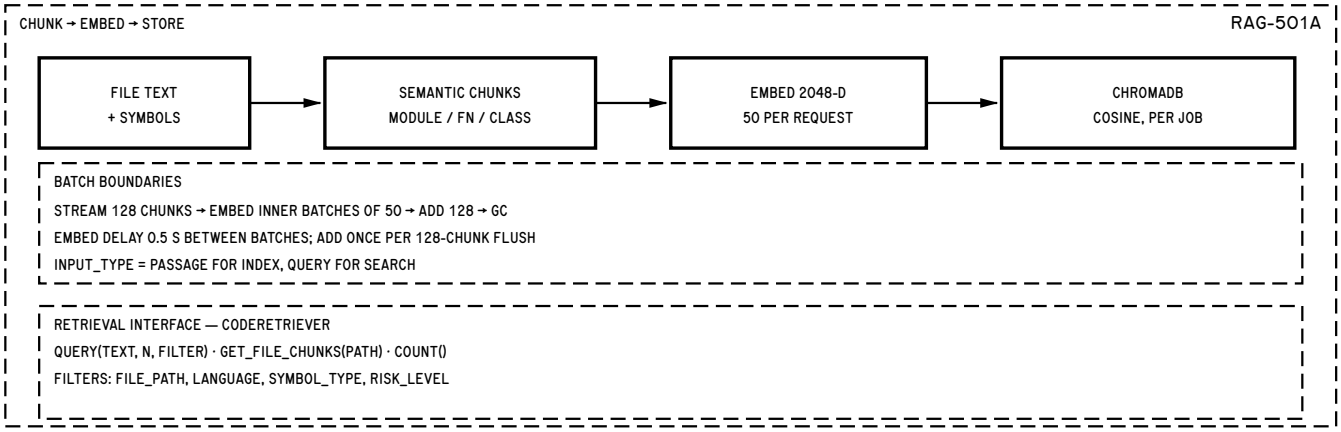


Figure 3.5-1. Semantic Chunking, Embedding and Retrieval Interface

3.5.2 FUNCTIONAL DESCRIPTION.

run names the collection *gnosis_{owner}_{repo}_{job_id}*, sanitised and length-capped, then gets or creates it with cosine distance. In a non-incremental run the collection is deleted first, so a re-run never mixes old vectors. For each file with chunks, it appends to a 128-chunk buffer; when the buffer fills, it requests embeddings, disambiguates duplicate identifiers, and adds documents with metadata. The final partial buffer follows the same path. Raw text is then released from state.

3.5.2.1 Dimension Guard.

On an incremental run, the stage peeks at one stored embedding. If its length differs from 2048, the collection is deleted and every file is re-embedded. This prevents a collection from mixing vectors produced by different models, which ChromaDB cannot query coherently. A changed embedding model therefore costs a full re-embed, not a partial one.

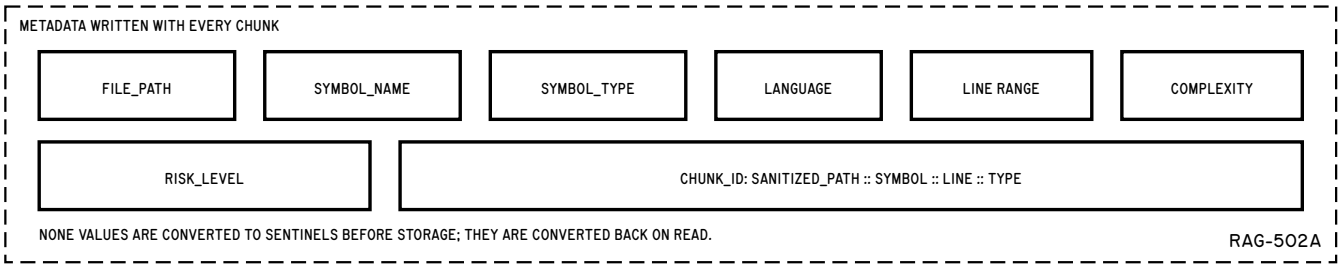


Figure 3.5-2. Chunk Metadata Contract

3.5.3 PERFORMANCE AND DESIGN DATA.

Table 3.5-1. Retrieval Parameters

Parameter	Value
Embedding model	nvidia/nemotron-3-embed-1b
Dimensions	2048
Embedding batch	50 texts
Stream buffer	128 chunks
Chroma add batch	128-chunk flush (500 constant unused)
Distance	cosine
Persistence	./chroma_db
Collection scope	owner + repo + job id

CAUTION

This stage clears *state.raw_contents* at the end. Stage 6's fallback therefore receives an empty string for every file, and explanation depends on retrieval succeeding. If the collection is empty or the retriever cannot connect, explanation degrades to a no-code prompt. See 8.1-1.

INDEX CHECK **MAINTAINER**

```
1 READ - collection name in state
2 CALL - retriever.count()
  √ COUNT - greater than zero
3 QUERY - a known function name
4 CONFIRM - file_path metadata matches
```

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